Synergy ERP — Local Setup Guide (Windows Native)

Complete step-by-step procedure from fresh git clone to running application on local PostgreSQL

1. Prerequisites Check

Verify that the required runtimes and tools are installed and present in your system PATH:

- **Node.js:** v22.x or v24.x (verify with: node -v)
- **npm:** v9.x or v11.x (verify with: npm -v)
- **Git:** Git for Windows (verify with: git -v)
- **PostgreSQL 18:** Running as a Windows Service (postgresql-x64-18) on Port 5000 with user postgres.

2. Clone Repository & Setup Branch

Open PowerShell, clone the project, and switch to your designated development branch:

```
git clone https://github.com/Aelyx-Work/synergy-erp.git
cd synergy-erp
git switch assign/akash
```

Prevent Git from pausing or waiting for an editor tab during future git pull operations:

```
git config core.mergeoptions "--no-edit"
```

3. Install Dependencies

Install monorepo dependencies across all workspaces (apps/web, services/api, packages, database):

```
npm install
```

4. Create Local Database in PostgreSQL

Using psql.exe on port 5000, create the synergy_erp database:

```
& "C:\Program Files\PostgreSQL\18\bin\psql.exe" -U postgres -p 5000 -c "CREATE DATABASE
synergy_erp;"
```

[NOTE] Enter your PostgreSQL master password when prompted (e.g. Akash.2003).

5. Environment Configuration (.env)

Create .env in the root project folder (c:\synergy-erp\.env). Key local values:

```
NODE_ENV=development
PORT=5001
API_PORT=5001
WEB_PORT=3000

POSTGRES_USER=postgres
POSTGRES_PASSWORD=Akash.2003
POSTGRES_DB=synergy_erp
POSTGRES_PORT=5000

DATABASE_URL="postgresql://postgres:Akash.2003@localhost:5000/synergy_erp?schema=public&connection_limit=20"
DIRECT_URL="postgresql://postgres:Akash.2003@localhost:5000/synergy_erp?schema=public"

NEXT_PUBLIC_API_URL=http://localhost:5001/api/v1
NEXT_PUBLIC_APP_URL=http://localhost:3000
```

[PORT ISOLATION] Because your local PostgreSQL service uses port 5000, the API is configured on port 5001 to prevent address collisions.

Frontend .env.local

Create apps/web/.env.local so Next.js targets port 5001 directly:

```
NEXT_PUBLIC_API_URL=http://localhost:5001/api/v1
NEXT_PUBLIC_APP_URL=http://localhost:3000
```

6. Build Packages & Generate Prisma Client

Compile shared TypeScript packages and generate the Prisma ORM client:

```
npm run build --workspace=@synergy/types
npm run build --workspace=@synergy/ui
npm run build --workspace=@synergy/validation
npm run db:generate
```

7. Apply Migrations & Synchronize Tables

Apply all 73 schema migrations and push any remaining unmigrated models (assembly batches, finished goods receipts):

```
npm run db:deploy
npx dotenv-cli -e .env -- npx prisma db push --schema=database/prisma/schema.prisma --accept-data-loss
```

Verify that migration state is fully synchronized:

```
npx dotenv-cli -e .env -- npx prisma migrate status --schema=database/prisma/schema.prisma
```

[SUCCESS] The output should confirm: "Database schema is up to date!".

8. Seed Master Data & Accounts

Populate the database with all 11 modules (21,135 HSN codes, materials, products, opening inventory, routes, BOMs):

```
npm run db:seed-all --workspace=@synergy/database
npm run db:seed-accounts-rbac --workspace=@synergy/database
npm run db:seed-accounts-test-user --workspace=@synergy/database
```

9. Launch Development Servers

> **[WINDOWS CAUTION]** Do NOT run "npm run dev" directly because it calls bash scripts/dev-local.sh (requires Docker/WSL). Always run "npm run dev:turbo".

```
npm run dev:turbo
```

Turborepo will build and start both the backend and frontend in parallel:

- **Web Application:** http://localhost:3000
- **Backend API:** http://localhost:5001 (Health check: http://localhost:5001/api/v1/health)

10. Login Credentials

Navigate to http://localhost:3000 in your browser and log in with any seeded account:

Role	Email	Password
Super Admin	admin@gmail.com	password123
Super Admin (Alt)	admin@synergy.com	password123
Store Manager	manager@synergy.com	password123
Sales Executive	sales@synergy.com	password123
Laser Cutting Head	laser.head@synergy.com	password123
Production Head	production.head@synergy.com	password123
Accounts / Finance	accounts.test@synergy.local	SynwTzRNbS_oXi0gPmNf1O03FoU!

11. Troubleshooting Windows Gotchas

- **Port Collision:** If port 3000 or 5001 is held by an old terminal, run: Get-NetTCPConnection -LocalPort 3000, 5001 | Stop-Process -Id {$_.OwningProcess} -Force
- **NODE_ENV Error:** services/api/package.json must use: "dev": "ts-node-dev --respawn --transpile-only src/server.ts" (without inline NODE_ENV=development which fails in cmd.exe).
- **Git Pull Stuck:** Run: git config core.mergeoptions "--no-edit" to prevent Git from waiting on editor tabs during pull.
- **401 on /events/stream:** Completely normal before logging in; visiting /login and authenticating with password123 resolves it.